

Hospital Data Analysis Project

```
CREATE TABLE Hospital1(Hospital_Name VARCHAR(100),  
Location VARCHAR(100),  
Department VARCHAR(100),  
Doctors_Count INT,  
Patients_Count INT,  
Admission_Date DATE,  
Discharge_Date DATE,  
Medical_Expenses DECIMAL(10,2)  
);
```

```
SELECT * FROM Hospital1;
```

1. Total Number of Patients

o Write an SQL query to find the total number of patients across all hospitals.

```
SELECT SUM(patients_count) AS Total_patients  
FROM Hospital1;
```

2. Average Number of Doctors per Hospital

o Retrieve the average count of doctors available in each hospital.

```
SELECT hospital_name, AVG(doctors_count) AS Average_Number_of_Doctors  
FROM Hospital1  
GROUP BY hospital_name;
```

3. Top 3 Departments with the Highest Number of Patients

o Find the top 3 hospital departments that have the highest number of patients.

```
SELECT department, SUM(patients_count) AS Total_patients
```

```
FROM Hospital1  
GROUP BY department  
ORDER BY Total_patients DESC  
LIMIT 3;
```

4. Hospital with the Maximum Medical Expenses

o Identify the hospital that recorded the highest medical expenses.

```
SELECT hospital_name, SUM(medical_expenses) AS Highest_medical_expense  
FROM hospital1  
GROUP BY hospital_name  
ORDER BY Highest_medical_expense DESC  
LIMIT 1;
```

5. Daily Average Medical Expenses

o Calculate the average medical expenses per day for each hospital.

```
SELECT hospital_name, SUM(medical_expenses)/SUM(discharge_date - admission_date) AS  
Average_medical_expenses_per_day  
FROM Hospital1  
WHERE discharge_date IS NOT NULL  
GROUP BY hospital_name;
```

6. Longest Hospital Stay

o Find the patient with the longest stay by calculating the difference between

Discharge Date and Admission Date.

```
SELECT *, (discharge_date - admission_date) AS Longest_patient_stay  
FROM Hospital1  
ORDER BY Longest_patient_stay DESC  
LIMIT 1;
```

7. Total Patients Treated Per City

- o Count the total number of patients treated in each city.

```
SELECT location AS City, SUM(patients_count) AS Total_number_of_patients
FROM Hospital1
GROUP BY location;
```

8. Average Length of Stay Per Department

- o Calculate the average number of days patients spend in each department.

```
SELECT department, AVG(discharge_date - admission_date) AS Average_Length_of_Stay
FROM Hospital1
WHERE discharge_date IS NOT NULL
GROUP BY department;
```

9. Identify the Department with the Lowest Number of Patients

- o Find the department with the least number of patients.

```
SELECT department, SUM(patients_count) AS Lowest_Number_of_Patients
FROM Hospital1
GROUP BY department
ORDER BY Lowest_Number_of_Patients ASC
LIMIT 1;
```

10. Monthly Medical Expenses Report

- Group the data by month and calculate the total medical expenses for each month.

```
SELECT
DATE_TRUNC('month', admission_date) AS Month,
SUM(medical_expenses) AS total_medical_expenses
```

FROM Hospital1

GROUP BY Month

ORDER BY Month;